

Lava / water / powder snow bucket desync when using it while rapidly turning

MC-12363 Networking

Status	Resolved / Fixed
Affected	Minecraft 1.5, Minecraft 1.6.1, Minecraft 1.6.2, Minecraft 1.7.4, Minecraft 14w27b, Minecraft 1.8 (+143 more)
Fixed in	Minecraft 14w30c, 21w11a
Reported	2013-03-19

STEPS TO REPRODUCE

1. Enter a world with a flat plain.
2. Give yourself a lava bucket.
3. Stand on the flat plain.
4. Move the mouse rapidly to spin around.
5. While spinning, repeatedly use the lava bucket to place the lava and pick it up again.
6. Observe that, about 1 in 10 attempts, the client and server desync: the bucket may appear full while lava is placed on the ground, or the bucket may appear empty while the lava was not placed and is still available to place.

OBSERVED BEHAVIOUR

Actual behavior

When the player rapidly changes their view direction while repeatedly using a lava, water, or powder snow bucket, the client and server can process the interaction using different target positions. This intermittently causes the client's predicted bucket state and world rendering to disagree with the server's state.

For example:

- The client may display the bucket as full even though the server placed the lava, water, or powder snow and the bucket is actually empty.
- The client may display the bucket as empty and show no placed fluid or powder snow, even though the server still considers the bucket full and allows the material to be placed.
- The client may simultaneously display the placed material and a full bucket.

The discrepancy is usually corrected after another bucket interaction or after the item is moved or otherwise updated in the inventory. Until then, the client-side inventory and world state do not accurately reflect the server-side state.

EXPECTED BEHAVIOUR

When using lava, water, or powder snow buckets while rapidly turning, each placement or pickup should be processed consistently, with the world and inventory showing the same result on both client and server. After placing a liquid or powder snow, the bucket should appear empty; after retrieving it, the bucket should appear full, without stale visuals or incorrect contents that require another action to update.

ENVIRONMENT

Minecraft Java Edition Versions: 1.5, 1.6.1, 1.6.2, 1.7.4, 14w27b, 1.8, 1.8.1-pre3, 1.8.8, 15w31c, 15w35e, 15w43a, 15w46a, 1.10.2, 16w42a, 1.12, 1.12.1 Pre-Release 1, 1.12.1, 1.12.2, 18w03b, 18w10d, 18w20c, 1.13-pre2, 1.13-pre3, 1.13-pre8, 1.13-pre9, 1.13-pre10, 1.13, 18w30b, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1, 1.13.2-pre1, 1.13.2-pre2, 1.13.2, 18w43b, 18w45a, 18w46a, 18w49a, 19w02a, 19w03a, 19w05a, 19w06a, 19w07a, 19w08a, 19w09a, 19w11a, 19w11b, 19w12b, 19w13a, 19w13b, 19w14a, 19w14b, 1.14 Pre-Release 1, 1.14 Pre-Release 2, 1.14 Pre-Release 3, 1.14 Pre-Release 4, 1.14 Pre-Release 5, 1.14, 1.14.1, 1.14.2 Pre-Release 1, 1.14.2 Pre-Release 2, 1.14.2, 1.14.3 Pre-Release 2, 1.14.3, 1.14.4 Pre-Release 1, 1.14.4 Pre-Release 3, 1.14.4 Pre-Release 4, 1.14.4 Pre-Release 5, 1.14.4 Pre-Release 6, 1.14.4, 19w34a, 19w35a, 19w36a, 19w37a, 19w38b, 19w39a, 19w40a, 19w41a, 19w42a, 19w45b, 19w46b, 1.15 Pre-release 1, 1.15 Pre-Release 2, 1.15 Pre-release 3, 1.15 Pre-release 4, 1.15 Pre-release 5, 1.15 Pre-release 6, 1.15 Pre-release 7, 1.15, 1.15.1, 1.15.1 Pre-release 1, 1.15.2 Pre-Release 1, 1.15.2 Pre-release 2, 1.15.2, 20w06a, 20w07a, 20w08a, 20w09a, 20w10a, 20w11a, 20w12a,

20w13a, 20w13b, 20w14a, 20w15a, 20w17a, 20w18a, 20w19a, 20w20a, 20w20b, 20w21a, 20w22a, 1.16 Pre-release 1, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16 Pre-release 5, 1.16 Pre-release 6, 1.16 Pre-release 7, 1.16 Pre-release 8, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w28a, 20w30a, 1.16.2 Pre-release 1, 1.16.2 Release Candidate 1, 1.16.2 Release Candidate 2, 1.16.2, 1.16.3 Release Candidate 1, 1.16.3, 1.16.4 Pre-release 1, 1.16.4 Pre-release 2, 1.16.4 Release Candidate 1, 1.16.4, 20w45a, 20w46a, 20w48a, 20w49a, 20w51a, 21w03a, 1.16.5, 21w05a, 21w05b, 21w06a, 21w07a, 21w08b, 21w10a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

A desync can happen when rapidly turning and trying to use a lava, water, or powder snow bucket. The bucket might appear full but you placed lava / water / powder snow; or the bucket might appear empty but you did not place lava / water / powder snow.

How to reproduce

- Give yourself a lava bucket
 - Spin around (move mouse) standing on a flat plain and repeatedly try to put down and pick up the lava in the bucket
- About 1 in 10 times, the client and server desync (e.g. I see the lava on the ground, and also in my bucket, until I use the bucket again or move it in my inventory; or I see no lava on the ground and my inventory displays the bucket as empty, but I actually still have lava and can place it).

Code analysis

Based on 1.12.2 decompiled using MCP 9.40

Using a bucket calls server-side `net.minecraft.item.ItemBucket.onItemRightClick(World, EntityPlayer, EnumHand)` which calculates the position itself and therefore can cause a desync. Ideally the position for picking up liquids and placing them should be calculated client-side only and then the server should only validate it and use that position.

Examples

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-12363>